

REQUIREMENTS BASELINE

Ollama-Based Customer-Service Chatbot

A private-by-design, web-first assistant for e-commerce support - grounded in approved content, connected to deterministic commerce APIs, and designed for safe human escalation.

EXECUTIVE INTENT Resolve routine product, policy, and authenticated order-status questions while preserving customer privacy and agent control. The model never writes directly to commerce data; every business operation passes through an authorized, schema-validated tool.

DOCUMENT CONTROL
Status Planning baseline
Audience Product, Architecture, Engineering, Security
Decision Approve scope and Phase 0 discovery

1. SCOPE AND MEASURABLE OUTCOMES				MVP boundary
SELF-SERVICE	TRUST	PRIVACY	OPERABILITY	
Product, policy, FAQ and authenticated order-status support.	Citations, abstention, policy controls and agent handoff.	Local inference, least-privilege tools and configurable retention.	Health, traces, quality evaluation, versioning and rollback.	

IN SCOPE Responsive web chat; English; RAG over approved policies/catalog content; authenticated read-only order tools; feedback; handoff; role-based knowledge publishing. **DEFERRED** Payments, refunds, order mutation, voice/social channels, multilingual support, autonomous outbound actions and model fine-tuning.

2. FUNCTIONAL REQUIREMENTS				Must-have unless marked
ID	CAPABILITY	TESTABLE REQUIREMENT		PRIORITY
FR-01	Customer experience	Embed a responsive Angular chat; stream responses; show sources, feedback and handoff; meet keyboard and screen-reader expectations.		MUST
FR-02	Session + identity	Support guest and authenticated sessions, expiry and correlation IDs; bind customer identity only after verified authentication.		MUST
FR-03	Routing + state	Route each turn to FAQ/RAG, commerce tool or human handoff using configurable policy and confidence thresholds; preserve bounded context.		MUST
FR-04	Grounded answers	Ingest approved knowledge with metadata and versions; retrieve, rank and cite evidence; abstain when supporting evidence is insufficient.		MUST
FR-05	Commerce tools	Call allowlisted catalog and order-status APIs through typed schemas with authorization, timeout, retry, idempotency and PII masking controls.		MUST
FR-06	Safety + privacy	Screen prompt injection, enforce allow/deny policy, redact secrets/PII from logs, apply retention rules and audit privileged changes.		MUST
FR-07	Human handoff	Create or enrich a support ticket with user-visible transcript, sources, identity context and reason for escalation - never hidden reasoning.		MUST
FR-08	Administration + ops	Provide role-based publish/rollback, model/prompt configuration, health checks, metrics, traces, alerts and an evaluation dashboard.		MUST

3. NON-FUNCTIONAL ACCEPTANCE TARGETS			Validate on target hardware
SERVICE 99.5% application availability target during the pilot service window. SECURITY TLS 1.2+, RBAC, managed secrets and audit coverage for privileged changes.	LATENCY p95 first token <= 4 s and complete answer <= 15 s at agreed pilot concurrency. PRIVACY No unnecessary raw PII in prompts/logs; 30-day transcript retention default, configurable.	QUALITY >= 90% citation coverage for knowledge answers; unauthorized tool actions = 0. RECOVERY Configuration/index RPO <= 24 h and RTO <= 4 h; model and prompt rollback tested.	

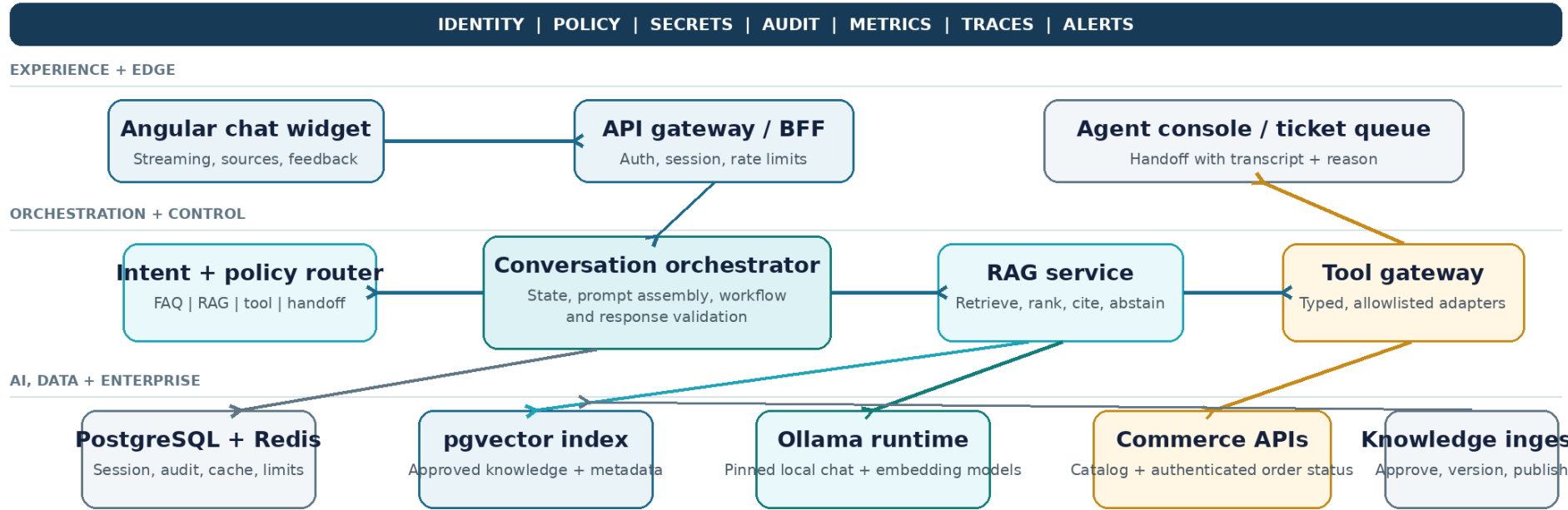
SOLUTION DESIGN

Reference Architecture and Major Components

Modular services in containers: Docker Compose for the pilot; Kubernetes/OpenShift when scaling or isolation requirements justify it.

4. LOGICAL ARCHITECTURE

Trust controls apply end to end



01	02	03	04	05	06
Authenticate + normalize	Classify + select route	Retrieve or call tool	Assemble prompt + infer	Validate, cite or abstain	Respond, observe or hand off

5. COMPONENT RESPONSIBILITIES AND DESIGN DECISIONS

FR traceability

COMPONENT	RESPONSIBILITY	KEY DESIGN DECISION	MAP
Web widget + BFF	Angular UI; REST/SSE boundary; session token; input limits and rate limiting.	Keep channel logic out of orchestration; no customer credentials in browser storage.	FR-01/02
Conversation orchestrator	State machine, intent/policy route, prompt assembly, tool loop and final response validation.	Deterministic workflow owns control; model proposes but cannot self-authorize.	FR-02/03/06
RAG + ingestion	Parse, chunk, embed, index, retrieve, rerank and cite approved content; publish/rollback versions.	Filter by tenant, locale, product and effective date; abstain below evidence threshold.	FR-04/08
Tool gateway	Typed adapters for catalog, order status and ticketing with timeouts, retries and circuit breakers.	Allowlist only; validate input/output schemas; recheck authorization per call; idempotency keys.	FR-05/07
Ollama runtime	Local streaming chat and embedding inference with health, resource limits and pinned model artifacts.	Separate chat/embedding models; benchmark quantization; controlled upgrade and rollback.	FR-03/04/08
Data, trust + operations	PostgreSQL/Redis/pgvector state; guardrails, handoff, OpenTelemetry traces, dashboards, alerts, evaluations and runbooks.	Encrypt and expire sensitive state; trace all calls; never log hidden reasoning; degrade safely to agent/FAQ.	FR-02/04/06/07/08

SECURITY BOUNDARY Retrieved text and model output are untrusted. Only authenticated application services may invoke allowlisted tools; the model has no direct database, shell or network access.

DELIVERY PLAN

Implementation Phases, Acceptance and Assumptions

Indicative MVP duration: 9-10 weeks with a small cross-functional team, subject to API readiness, content quality and target-hardware benchmarks.

6. PHASED IMPLEMENTATION PLAN				Each phase ends at a usable gate
PHASE	FOCUS	TIME	PRIMARY OUTPUT	EXIT GATE
0	Align + baseline	1 wk	Architecture decisions, data classification, source inventory, API contracts, test set and delivery backlog.	Scope, SLOs, owners and risks approved.
1	Vertical slice	2 wk	Angular widget, BFF, session flow, orchestrator, Ollama streaming, health checks and basic FAQ route.	End-to-end answer and session flow in test.
2	Knowledge grounding	2 wk	Ingestion, pgvector, metadata filters, citations, abstention, content publishing and evaluation harness.	Grounding targets pass on agreed test set.
3	Commerce + handoff	2-3 wk	Catalog/order-status adapters, identity binding, typed tools, PII controls and support-ticket transfer.	Read-only authorized flows and negative tests pass.
4	Hardening + pilot	2 wk	Security, load, resilience, observability, backup/restore, runbooks, user acceptance and rollout controls.	Go-live checklist, dashboard and rollback drill approved.
5	Production evolution	Ongoing	Feedback tuning, content expansion, GPU/horizontal scale, new channels and separately approved actions.	Each change has its own safety and acceptance gate.

7. VERIFICATION AND RELEASE ACCEPTANCE			Evidence required
FUNCTIONAL FR-01 through FR-08 traced to automated UI/API tests and signed product acceptance.	GROUNDING Agreed curated questions meet citation and correctness targets; unsupported questions abstain safely.	SECURITY Prompt-injection, broken authorization, data leakage, abuse and dependency scans pass with no critical issue.	
PERFORMANCE Latency, capacity, error rate and resource use pass under agreed pilot load on target hardware.	RESILIENCE API/model timeout, circuit-breaker, handoff, backup/restore and model/prompt rollback are demonstrated.	OPERATIONS Dashboards, alerts, audit trail, content ownership, incident runbook and support handover are approved.	

8. KEY ASSUMPTIONS
A1 A Dell PowerEdge R630-class host with about 64 GB RAM is the CPU-first pilot baseline; capacity must be benchmarked before the SLO is committed.
A2 Approved policy/product content has named owners; catalog, order-status and ticket APIs are documented and reachable.
A3 MVP order access is authenticated and read-only; guest users cannot retrieve customer-specific information.
A4 English and responsive web are first; WCAG 2.1 AA is the accessibility target; transcript retention defaults to 30 days, subject to policy.
A5 Model, prompt and index versions are pinned; no fine-tuning or autonomous commerce action is included in the first release.

9. PRIMARY RISKS AND CONTROLS
Latency / capacity Use a right-sized quantized model, streaming, caching, bounded queues and load-based scale decisions.
Hallucination / stale content Require evidence, cite sources, abstain below threshold, assign owners and evaluate each published version.
Prompt injection / tool abuse Treat retrieved text as untrusted; enforce typed allowlisted tools, per-call authorization and output scanning.
Integration outage Apply timeout, retry, circuit breaker and graceful degradation to FAQ or agent handoff.
Unclear operating ownership Name product, content, model, security and on-call owners before the pilot gate.

NEXT DECISION Approve the MVP boundary and authorize Phase 0 to validate model licensing, source ownership, API readiness and measured Ollama capacity.